

1. Semaphores

A semaphore is a variable used by the operating system to control and synchronize processes. It helps control access to a shared resource so that processes do not use it incorrectly at the same time.

Example

Suppose there is only one printer and many processes want to use it. A semaphore allows only one process to use the printer at a time. After that process finishes printing, the semaphore allows another process to use the printer.

Types of Semaphores

- **Binary Semaphore**

A binary semaphore has only two values: 0 and 1. It is used when only one process can access a resource at a time.

Example: One printer shared by many processes.

- **Counting Semaphore**

A counting semaphore can have values greater than 1. It is used when there are multiple copies of a resource.

Example: If there are 3 printers, the semaphore can allow 3 processes to use them at the same time.

2. Signals

A signal is a notification sent to a process to inform it that an event has occurred.

Example

When a program is running and the user presses Ctrl + C, the operating system sends a SIGINT signal to the program. The program can then stop or handle the interruption.

Common Examples of Signals

- **SIGINT** – Interrupts a running process.

- **SIGTERM** – Requests a process to terminate.
- **SIGKILL** – Forces a process to terminate.
- **SIGSTOP** – Stops a process.
- **SIGCONT** – Continues a stopped process.